We previously saw how to write sinusoids in phasor notation, which allowed us to much more easily add together multiple simple harmonic motions using vector concepts. Today we will extend this idea to:

Complex representation of waves:

Begin with a sinusoidal wave propagating to the right:

$$y(x,t) = A\cos(kx-\omega t+\phi_0)$$

and express the cosine as $\mathrm{Re}\left[e^{j(kx-\omega t+\phi_0)}\right]$ and define the complex-valued wave $\tilde{y}(x,t)$ as

$$\tilde{y}(x,t) = \underbrace{(Ae^{j\phi_0})}_{\text{complex amplitude}}\underbrace{e^{j(kx-\omega t)}}_{\text{unit vector rotating in the complex plane}}$$

so that $y(x,t) = \mathrm{Re}[\tilde{y}(x,t)]$

The complex amplitude is $\tilde{A} = Ae^{j\phi_0}$ as for phasors.

$\Rightarrow$ $|\tilde{A}|$ gives wave amplitude and phase of $\tilde{A}$ gives wave phase constant ϕ_0

⇒ Top Hat Question on slide

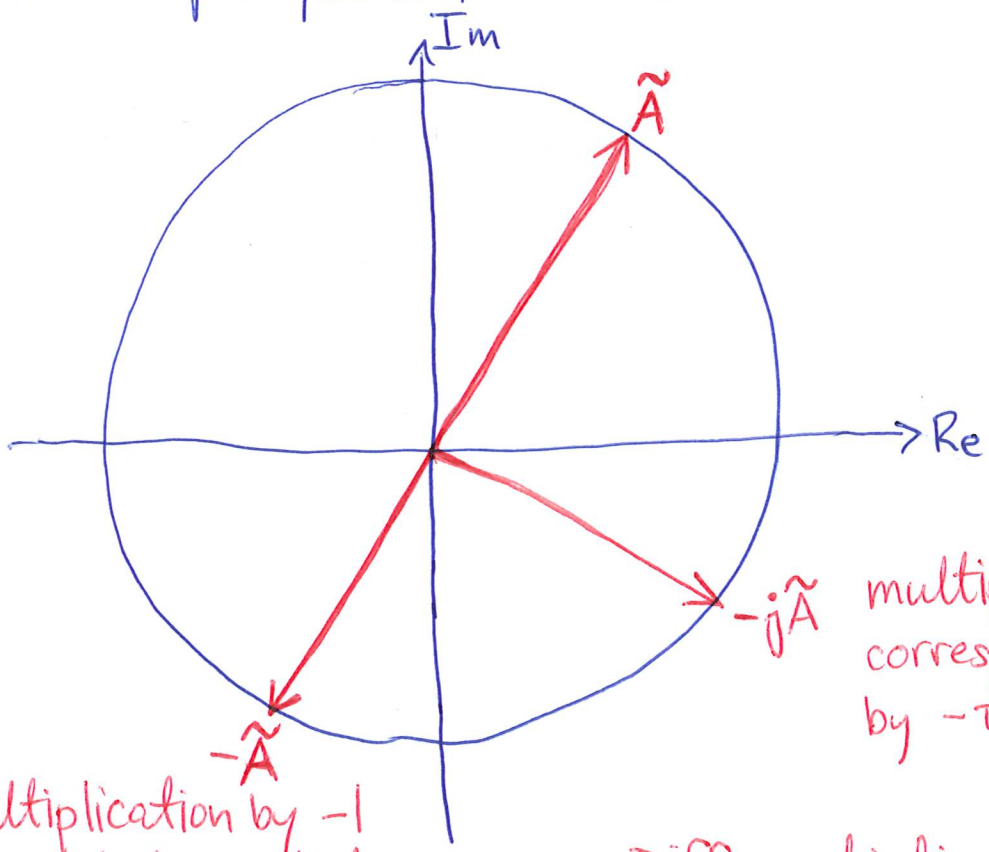
What do the complex representations of velocity and acceleration look like? (Note: these are of the particles)

$$\tilde{y} = \tilde{A} e^{j(kx-\omega t)}$$

$$\tilde{v}_y = \frac{\partial \tilde{y}}{\partial t} = -j\omega \tilde{A} e^{j(kx-\omega t)}$$

$$\tilde{a}_y = \frac{\partial^2 \tilde{y}}{\partial t^2} = (-j\omega)^2 \tilde{A} e^{j(kx-\omega t)} = -\omega^2 \tilde{A} e^{j(kx-\omega t)}$$

In the complex plane this looks like:



multiplication by $-j$ corresponds to a rotation by $-\pi/2$

multiplication by -1 corresponds to a rotation by $-\pi$

* Differentiation w.r.t. t leads to a rotation by $-\pi/2$.

This is another way to see why acceleration leads velocity, which leads position.

Example: two waves $y_1 = (2.0\text{m})\sin(kx - \omega t + 3.67)$

$y_2 = (2.0\text{m})\cos(kx - \omega t)$

travel together down a string with $\omega = 1$ rad/s. What is the resulting wave $y_3 = y_1 + y_2$ amplitude and phase constant?

Solution: Start by expressing y_1 and y_2 in complex form:

Need to convert y_1 to a cosine using

$$\sin(a \pm b) = \sin a \cos b \pm \sin b \cos a$$

will contain $kx - \omega t + \phi_1$.

chosen so that first term is zero and second term gives $\cos a$

$\Rightarrow \cos b = 0$ and $\sin b = 1$ with + sign

$\hookrightarrow b = \pi/2$

Therefore $y_1 = (2.0\text{m})\sin\left(\underbrace{[kx - \omega t + 3.67 - \pi/2]}_{a} + \underbrace{[\pi/2]}_{b}\right)$

$$\Rightarrow y_1 = (2.0\text{m})\cos\left(kx - \omega t + \underbrace{[3.67 - \pi/2]}_{\phi_1}\right)$$

and y_2 is already in the correct form with $\phi_2 = 0$, so

$$\tilde{y}_1 = (2.0\, e^{j\phi_1})\, e^{j(kx - \omega t)}$$

$$\tilde{y}_2 = (2.0)\, e^{j(kx - \omega t)}$$

Write the complex amplitudes in rectangular form:

$$\tilde{A}_1 = 2.0\cos\phi_1 + j2.0\sin\phi_1 = -1.008317 + j\,1.727222$$

$$\tilde{A}_2 = 2.0 + j(0)$$

Now add them together:

$$\tilde{A}_3 = \tilde{A}_1 + \tilde{A}_2 = 0.991683 + j\,1.727222$$

The amplitude of the resulting wave is:

$$|\tilde{A}| = \sqrt{0.991683^2 + 1.727222^2} = \boxed{1.99} \rightarrow 2.0 \text{ to 2 sig figs}$$

The real and imaginary parts are both + so this is in quadrant 1 and the phase constant is

$$\phi_3 = \tan^{-1}\left(\frac{1.727222}{0.991683}\right) = \boxed{1.05 \text{ rad}} \text{ or } 60.14^\circ$$

Beware: the textbook uses a bunch of trig identities to discuss wave interference but using complex representations is a much more straightforward approach. All of the results from section 16.5 can be obtained using complex representation without resorting to "equation hunting" and being unsure about whether a particular equation applies.

Let's now start looking at different media that waves are propagating in. Start with a stretched string:



what are some physical parameters of the string that could determine how fast a wave travels?

- linear mass density will dictate how much a small length of the string will respond to forces acting on it (inertia).
- tension in the string dictates the strength of forces acting on that length

$\Rightarrow$ that's it!

linear mass density μ has dimension $\frac{[M]}{[L]}$

tension T has dimension $\frac{[M][L]}{[T]^2}$

so $\frac{T}{\mu}$ has dimension $\frac{[M][L]}{[T]^2}\frac{[L]}{[M]} = \frac{[L]^2}{[T]^2}$

the same as a velocity (m/s) but squared.

$\Rightarrow$ the wave speed must be of the form $v = C\sqrt{\frac{T}{\mu}}$

where C is a constant. Actual analysis of the physics looking at forces and using Newton's laws yield $C=1$, so for a taut string:

$$\boxed{v=\sqrt{\frac{T}{\mu}}}$$ wave speed on a string.

Example: A 6.0m long rope has a mass of 1.0kg. It is held taut with a tension of 12N of force. How long does it take for a wave to travel from one end to the other?

Solution:

We need $\mu = \frac{1.0\,kg}{6.0\,m} = 0.167\,kg/m$, we have $T = 12N$

$$\Rightarrow v = \sqrt{\frac{12N}{0.167\,kg/m}} = 8.48\,m/s$$

The wave moves at constant speed, so

$$\Delta t = \frac{6.0m}{8.48\,m/s} = \boxed{0.71\,s}$$